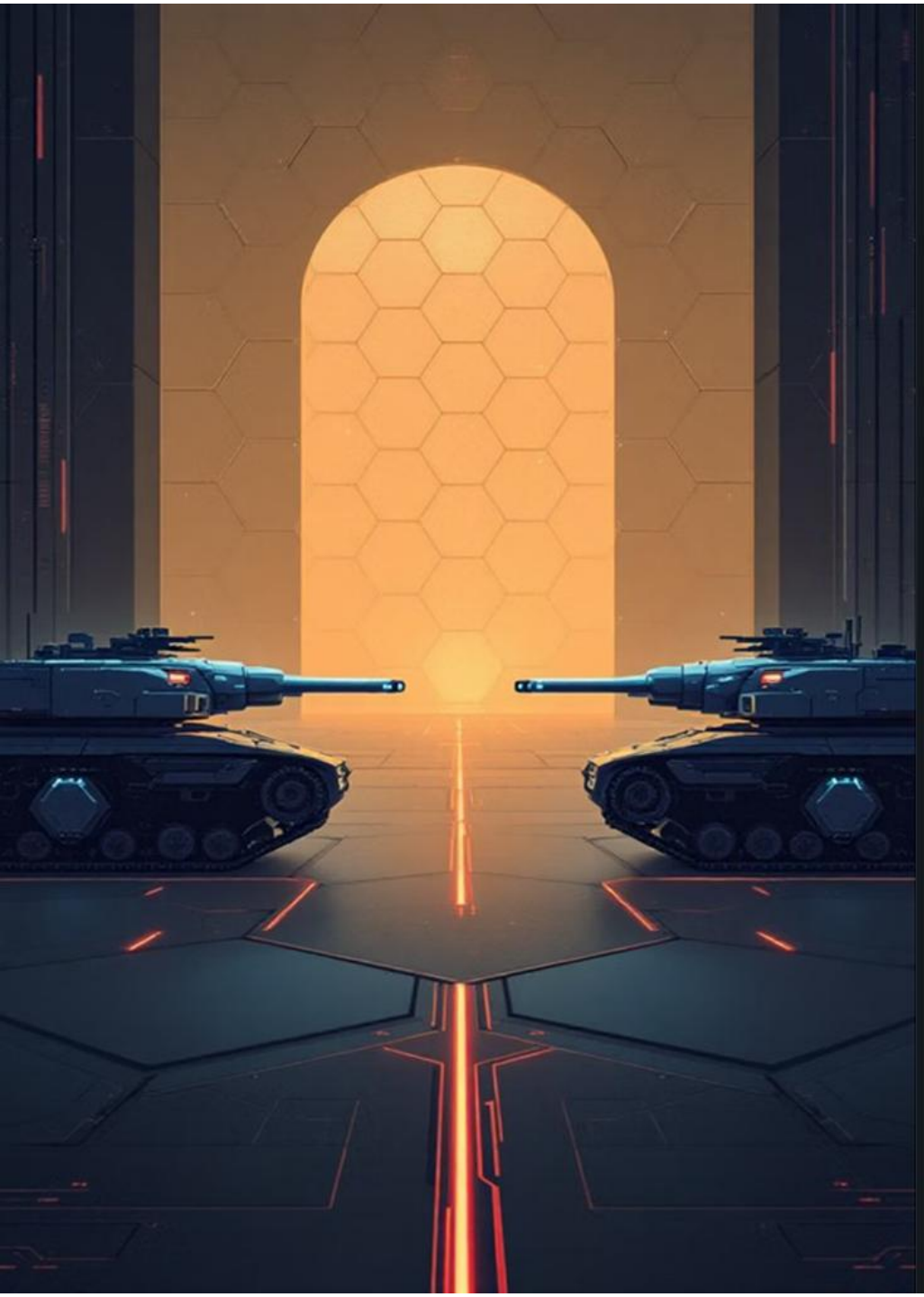

Hex Siege Arena

AI Tactical Game

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CSE 3210 AI Lab Project Proposal - Turn-based Tactical Combat Game on a Hexagonal Grid

TACTICAL OVERVIEW: MECHANICS AND IN HEX TANKS STRATEGY

A deconstruction of turn-based armor combat, terrain management, and win-state execution.

DYNAMIC STATE MODIFIERS (POWER-UPS)

ATTACK MULTIPLIER

Mechanic: Doubles damage output.

The Math: Base DMG * 2

(Example: Ktank 3 DMG -> 6 DMG)

SHIELD BUFFER

Mechanic: Absolute damage negation.

JetBrains Mono

The Math: Blocks exactly 2 hit instances, regardless of damage value.

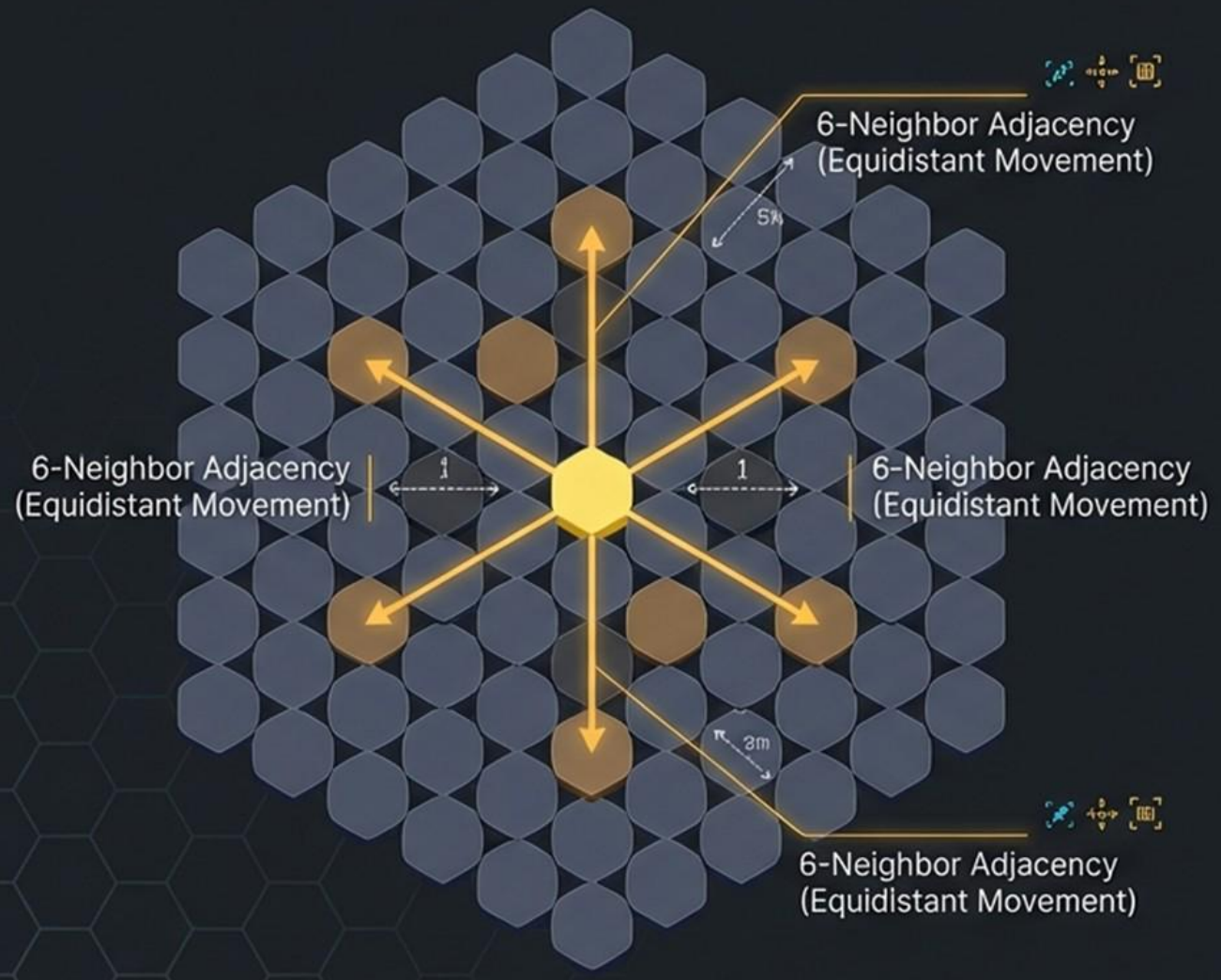
BONUS MOVE

Mechanic: Action Chaining.

The Math: Actions = Actions + 1

(Temporarily breaks the Mutual Exclusion rule)

HEXAGONAL TOPE TOPOLOGY & BOARD STATE



GRID SPECIFICATIONS

- **Total Cells:** 91
- **Topology:** Flat-topped Hexagons
- **Center Coordinate:** (0, 0)

TERRAIN CLASSIFICATION

1. Indestructible Walls

Static constraints (Dark Grey).
Cannot be moved through or destroyed.

2. Destructible Blocks

Dynamic cover.
Brown (HP: 2) | Silver (HP: 3).

3. Power Tiles

Modifier zones (Purple/HP: 2). Grants buffs upon destruction.

How Q Tank Moves

UNIT PROFILE: QTANK (THE SNIPER)

ROLE: SPATIAL CONTROL

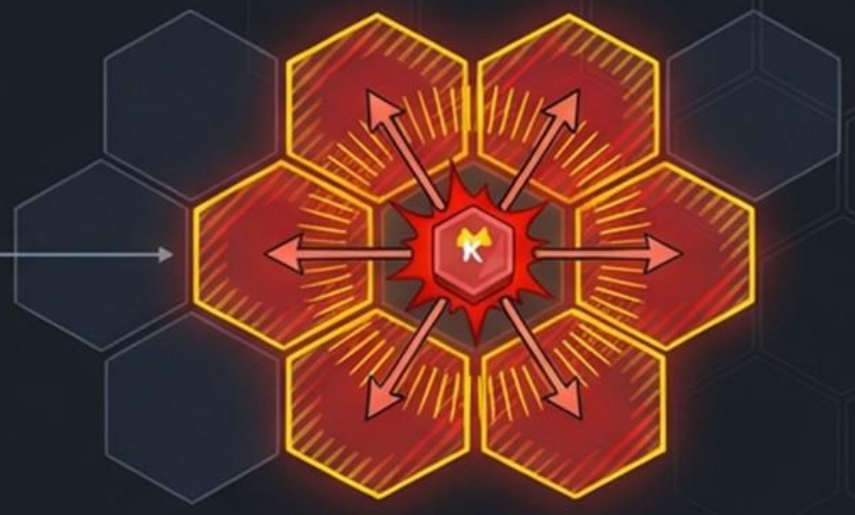
- **Health:** 8 HP
- **Mobility:** High

ATTACK LOGIC: RAYCASTING

- **Method:** Projects damage vector along axial directions.
- **Constraint:** Projectile terminates immediately upon first collision (Unit, Wall, or Block).
- **Output:** 2 Damage per hit.

UNIT PROFILE: KTANK (THE JUGGERNAUT)

ATTACK LOGIC: AREA OF EFFECT (AOE)



ROLE: THE KING (HIGH VALUE TARGET)

- **Health:** 10 HP
- **Mobility:** Low (1 Hex/Turn)

ATTACK LOGIC: AREA OF EFFECT (AOE)

- **Method:** Radial explosion affecting the unit's cell + all 6 neighbors.
- **Output:** 3 Damage (High Impact).
- **Tactical Constraint:** FRIENDLY FIRE ENABLED. The AI must calculate risk vs. reward before firing.

VICTORY CONDITIONS & LOOP TERMINATION

CONDITION A: REGICIDE (KING-KILL)

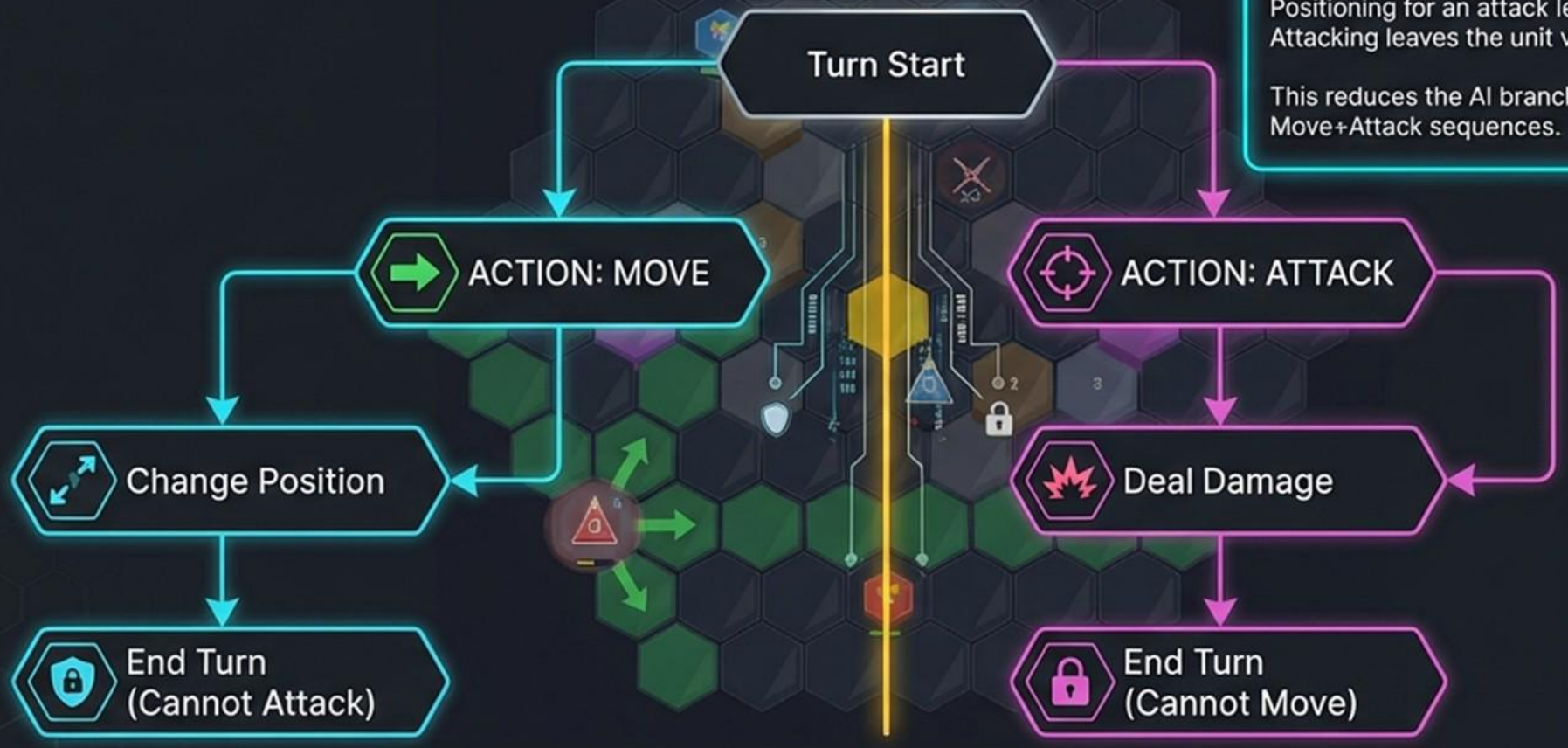
- **Trigger:** Enemy King HP <= 0.
- **Result:** Immediate Termination.

CONDITION B: CENTER CONTROL

- **Trigger:** King Unit occupies Coordinate `(0,0)`.
- **Constraint:** Must hold position until the start of the NEXT turn.
- **Design Intent:** Acts as a 'Forcing Function' to prevent camping and draw states by concentrating conflict at the board center.

THE ACTION ECONOMY

THE MUTUAL EXCLUSION RULE

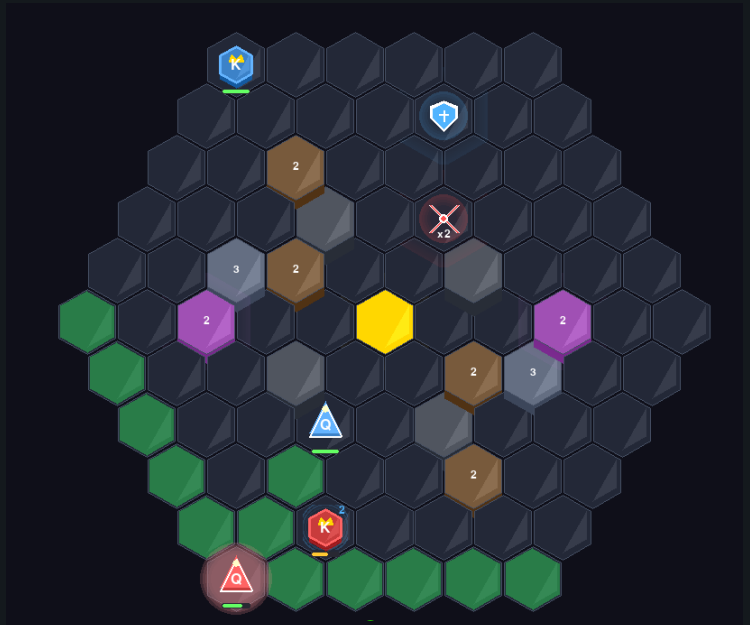


STRATEGIC IMPLICATION

Opportunity Cost: Every decision requires a sacrifice. Positioning for an attack leaves the unit stationary. Attacking leaves the unit vulnerable to retaliation.

This reduces the AI branching factor by eliminating Move+Attack sequences.

Intermiditate States



Thank You
Any question?